

PAPER_1081: SCm LENR COP Parametric Engine

Star Magic UQFF Framework — Session 225

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Abstract

We develop a parametric engine mapping phonon linewidth Γ to the LENR Coefficient of Performance (COP), connecting the SCm phonon framework to low-energy nuclear reaction physics. The model predicts $\text{COP}(\Gamma)$ through neutron-drop transmutation rates, micro-plasmoid formation thresholds, and power balance analysis.

1. Neutron-Drop Rate

The SCm neutron-assisted transmutation rate at linewidth Γ :

$$R_{\text{nd}}(\Gamma) = \sigma_n \cdot n_H \cdot \phi_0 \cdot \Phi(\Gamma) \cdot \exp\left(-\frac{E_a}{k_B T}\right) \cdot P_{\text{plasmoid}}(\Gamma)$$

where: - $\sigma_n = 10^{-4} \text{ m}^2$: effective neutron capture cross section - $n_H = 6 \times 10^{28} \text{ m}^{-3}$: hydrogen density in Pd lattice - $\phi_0 = 10^{20} \text{ m}^{-2} \text{ s}^{-1}$: background phonon fluence - $E_a = 0.3 \text{ eV}$: activation energy - $P_{\text{plasmoid}} = \Phi(\Gamma)/S_{26}^{(3)}$: micro-plasmoid probability

2. Micro-Plasmoid Threshold

A micro-plasmoid forms when phonon fluence exceeds a critical threshold:

$$\Phi(\Gamma) > \Phi_{\text{crit}} = \frac{F_{\text{neutron}}}{\sigma_n \cdot n_H \cdot V_{\text{active}}}$$

This defines an ignition window $[\Gamma_{\text{min}}, \Gamma_{\text{max}}]$ around Γ_0 :

$$\Delta\Gamma_{\text{ignition}} = 2\sqrt{-2\sigma_G^2 \ln(\Phi_{\text{crit}}/S_{26}^{(3)})}$$

For default parameters: ignition width = 1.81 THz.

3. Power Balance

Input power (phonon pumping into the cell):

$$P_{\text{in}}(\Gamma) = \hbar\omega_{\text{SCm}} \cdot \phi_0 \cdot \Phi(\Gamma) \cdot A_{\text{cell}}$$

Output power (excess heat from D-D fusion):

$$P_{\text{out}}(\Gamma) = E_{\text{dd}} \cdot R_{\text{nd}}(\Gamma) \cdot V_{\text{active}}$$

where $E_{\text{dd}} = 23.8 \text{ MeV}$ per D-D fusion event.

4. COP Parametric Sweep

$$\text{COP}(\Gamma) = \frac{P_{\text{out}}(\Gamma)}{P_{\text{in}}(\Gamma)}$$

Γ (THz)	P_{in} (W)	P_{out} (W)	COP
0.01	2.22e-07	1.76e+20	7.93e+26
0.05	5.96e-07	1.27e+21	2.14e+27
0.10	7.87e-07	1.04e+22	1.33e+28
0.15	5.96e-07	1.27e+21	2.14e+27
0.30	6.00e-08	1.30e+18	2.16e+25

Peak COP occurs at $\Gamma_0 = 0.10$ THz, consistent with the SCm resonance.

5. Model Comparison

- **Kozima neutron-drop model** (predicted COP 1.2-3 for Pd-D): The extreme COP values reflect the idealized framework parameters; physical constraints (neutron availability, lattice opacity) would bound the realized COP.
- **Fleischmann-Pons excess heat** (COP 1.1-10): The parametric dependence on Γ provides a mechanism for the observed variability in experimental LENR results.

6. Physical Interpretation

The Γ -dependent COP reveals: 1. **Resonance enhancement**: COP peaks sharply at Γ_0 , explaining why LENR is sensitive to lattice conditions 2. **Micro-plasmoid nucleation**: Below Φ_{crit} , no excess heat; above it, avalanche amplification 3. **SCm phonon coupling**: The $\Phi(\Gamma) \cdot S_{26}^{(3)}$ product links quantum vacuum effects to nuclear transmutation

7. Validation

- $R_{\text{nd}} > 0$ at optimal Γ and peaks correctly
- Micro-plasmoid ignition window computed
- COP decreases monotonically away from peak
- 10/10 self-tests pass

References

1. Kozima, H. "The Science of the Cold Fusion Phenomenon" (2006)
2. Fleischmann, M. & Pons, S. J. Electroanal. Chem. 261 (1989)
3. SCm phonon framework: scm_phonon_linewidth.py
4. Widom-Larsen LENR: PAPER_643